Diesel Filter Modernization: Predictive Maintenance and Intelligent Fault Diagnosis Using Machine Learning and Data Analytics

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***Abstract*—**Diesel particulate filters (DPFs) and fuel filtering units are essential emission abatement and engine protection systems that deteriorate under diverse real-world operating conditions resulting in unplanned maintenance activities, non-compliance, and premature wear of the engine. Time and interval-based filter maintenance schedules do not reflect the complex nonlinear nature of soot and ash loading, and the onset of bypass failure in a diverse fleet. This research paper explores in depth the use of machine learning (ML) and data analytics to digitally transform diesel filter maintenance. Data from sensors measuring differential pressure, exhaust gas temperature, engine load, fuel flow, and a calculated soot accumulation index were gathered from 42 diesel trucks and buses operating in northern India over 18 months. Random Forest, XGBoost, Long Short-Term Memory (LSTM), Support Vector Machine (SVM), and a hybrid Convolutional Neural Network-LSTM (CNN-LSTM) models were tested under a uniform experimental set-up. The CNNLSTM architecture provided the highest accuracy of 94.3% in predicting clogging of the DPF (F1-score 0.937), and XGBoost model best performed six-class classification of fuel filter faults at 91.8% accuracy. An optimized predictive maintenance schedule based on these models led to 34% fewer unnecessarily replaced filters and 61% fewer unexpected downtime events compared to the fleet's current fixedinterval maintenance protocol, with an average lead time of 47 hours. The paper also outlines an entire system architecture of edge-compatible filter health monitoring, and notes limitations, ethical considerations and future work including digital twin integration and federated fleet learning.

***Keywords*—**diesel particulate filter; predictive maintenance; machine learning; XGBoost; LSTM; fault diagnosis; data analytics; condition-based maintenance; DPF regeneration; soot accumulation

I. INTRODUCTION

For more than 100 years diesel engines have driven global trade, farming and industrial development, and remain vital for freight transport, construction, and mechanised farming - especially in emerging markets. Their torque density, efficiency, and durability mean that they are unlikely to be replaced by alternative powertrains in the short- to mid-term in most high duty-cycle workloads. But this ongoing significance is predicated on continued adherence to ever more stringent emission regulations: Euro VI in Europe, Bharat Stage VI (BS-VI) in India, EPA Tier 4 Final in North America. These standards make the diesel aftertreatment system - particularly the diesel particulate filter (DPF) - a critical component of engine design and fleet management.

The DPF is designed to retain particulate matter (PM) in engine exhaust before release, achieving greater than 99% PM2.5 reduction with optimal performance [1]. But the DPF is in a dynamic state: combustion soot deposition occurs within the porous cordierite or silicon carbide substrate of the DPF, causing backpressure and engine efficiency loss until the accumulated soot is removed (or regenerated), either passively by oxidation at high exhaust temperature (such as when driving on the highway) or actively by a thermal event triggered by fuel injection. Alongside soot, non-combustible ash from engine oil and fuel contaminants accumulate and become trapped, reducing filter effectiveness over time [2]. Fleet maintenance faces a core challenge in managing this complex, non-linear degradation process.

Conventional maintenance approaches respond to this challenge with either calendar-based or mileage-based service intervals, occasionally supplemented by dashboard warning lights triggered by threshold-level differential pressure events. Neither approach is satisfactory: arbitrary intervals cannot compensate for the enormous variability in filter loading caused by changes in duty cycle, fuel quality, engine condition, and other factors. A DPF operating in urban stopstart service may reach regeneration threshold in half the time required for the same filter on highway duty, yet both would trigger the same fixed-interval maintenance event. This results in a bimodal distribution of failures: premature maintenance on under-loaded filters in benign operating conditions, and missed or deferred maintenance on over-loaded filters in harsh operating conditions [3].

The availability of low cost on-board diagnostic (OBD-II) sensors, fleet telematics and robust machine learning (ML) techniques has opened the possibility of a new approach: data-driven predictive maintenance (PdM) that estimates filter state of health from on-vehicle sensor data and issues maintenance recommendations that take account of actual operating conditions rather than proximity to the calendar. This paper explores how this opportunity can be harnessed - which ML models are appropriate to which parts of the problem, how a deployable system can be designed, and what improvements in maintenance efficiency and failure avoidance can be achieved compared to traditional approaches.

A. Research Objectives

This research focuses on four questions:

1. Are ML models trained on multi-variate OBD sensor signals capable of accurately and timely predicting DPF clogging events in order to scheduled timely maintenance interventions in diverse fleets of commercial vehicles?

2. What is the optimal machine learning architecture to model the temporal evolution of soot loading and filter performance, and how does the choice of architecture affect the different statistical relationships present in DPF and fuel filter fault detection tasks?
3. What are the measurable maintenance efficiency gains (measured in terms of avoiding unnecessary replacements, reducing unplanned downtimes and advance warning) associated with ML-based PdM as compared to fixed-interval maintenance schedules?
4. How can an edge-deployable filter health monitoring system be designed to enable continuous ML inference on the limited computing and connectivity resources available in commercial vehicles?

II. LITERATURE REVIEW AND RELATED WORK

A. Physics-Based DPF Models and Their Limitations

The first physics-based models to describe DPF behavior were laid out in 1984 by Bisset with the first analytical model of thermal regeneration in wall-flow monolith filters [4]. Refinements by Konstandopoulos and Johnson [5] yielded the pressure drop model still used in most production DPF monitoring systems, which relates differential pressure and soot loading to a wallflow filtration equation that is parameterized by filter geometry and soot layer properties. While these physics-based models are effective in the laboratory, they do not translate well into practice due to the need for accurate knowledge of soot density, ash concentration and filter substrate properties that all change with fuel quality, engine efficiency and wear.

Efforts to apply physics-based models in practical applications have resulted in semi-empirical extensions recalibrating model parameters with periodic sensor measurements [3]. While such models are more accurate than fully open-loop physics-based models, they are still prone to sensor errors and require recalibration that introduces operational complexity. More importantly, physics-based models are constructed to predict a given physical quantity (such as soot mass or differential pressure) rather than to classify maintenance-relevant system conditions (which is important if the goal is to provide actionable maintenance decisions rather than estimates of the physical state).

B. Machine Learning Applications in Engine Fault Diagnosis

The use of ML for diesel engine diagnostics grew with the release of large automotive sensor data sets. Theissler et al. [6] used Random Forest classifiers on ECU sensor time series to diagnose multi-class automotive faults with 8-12% higher accuracy than threshold-based rules for various fault classes. Significantly, that study demonstrated that diagnostic signatures - derived from tree-based feature importance analysis - were physically interpretable, offering a pathway from data-driven solutions to engineering insight.

Zhao et al. [7] also explored LSTM models for component wear prediction from vibration time series, confirming that LSTM models with memory cells significantly outperformed static classifiers on time-series data with long time-dependency, which is relevant to the modeling of soot accumulation, which involves a slow degradation process over tens of hours to days. In more recent work, Gao et al. [8] used Bayesian neural networks for predicting the state of DPFs, including uncertainty modeling to improve the threshold for reporting anomalies in safety-critical monitoring applications.

In fuel systems, Wang et al. [9] showed that XGBoost could be used to classify six fuel contamination types in heavy-duty diesel systems with 93.1% accuracy using a feature set extracted from fuel pressure, flow rate and injector commands. The success of gradient-boosting ensemble methods in structured, tabular fault detection and classification tasks (as opposed to temporal, time series degradation estimation) is a common theme in the literature that guides model choice in this paper.

C. Predictive Maintenance Frameworks for Vehicles

There is a body of industrial and academic research on end-to-end fleet PdM. Mobley [10] offers the basic framework of reactive, preventive and predictive maintenance, laying the groundwork for condition-based monitoring. More recent research has addressed the challenges of applying ML-based PdM to vehicle fleets: the diversity of data across different manufacturers and models, the cost of ground truth for training ML algorithms for rare faults and the computational efficiency of inference in vehicle ECUs [11].

A persistent limitation in the published literature is the absence of controlled comparative evaluation across multiple ML architectures under identical experimental conditions, with performance measured against a quantified baseline of conventional maintenance practice. Most published PdM studies report classification accuracy in isolation without quantifying the downstream impact on maintenance scheduling efficiency. This paper addresses both gaps.

III. DIESEL FILTER SYSTEMS: OPERATING PRINCIPLES AND FAILURE MODES

A. Diesel Particulate Filter Operation

A DPF is a wall-flow filter that requires exhaust gas to flow through the walls of parallel channels, trapping particulate matter in the wall structure and on the upstream face of the channels. The DPF is operated in load-regeneration cycles. Loading involves soot build-up and an increase in pressure across the filter; when soot accumulation reaches a certain level (typically 8-12 g/L depending on filter size), regeneration occurs. Passive regeneration is achieved when high exhaust temperatures (greater than 550°C) are maintained for longer than a threshold time, converting the soot into CO_2. When passive regeneration conditions are not satisfied, active regeneration is electronically initiated, which involves injecting additional (post injection) fuel to increase exhaust temperatures [2].

The ash that results from engine oil combustion and metallic fuel additives does not oxidise during regeneration, and remains in the filter, reducing its effective volume and causing a rise in the base differential pressure (DP base) regardless of soot mass. Ash load is the key factor driving DPF lifetime, and requires filter removal for cleaning or replacement after 150,000-300,000 km, depending on engine oil consumption and fuel quality.

B. Fuel Filter Operation and Failure Taxonomy

Diesel fuel filters are essential to guard injection system components (high pressure pump, common rail injectors) from particulate contamination and water contamination. Current common rail systems inject at pressures of greater than 2,000 bar, making injection system components highly susceptible to contamination by particles greater than 4-10 microns. Failure of fuel filters is characterised by six primary modes of failure, which are the targets of classification in this paper: (1) typical wear/ageing/sediment build-up, (2) microbial contamination (biofouling), (3) water ingress and free water accumulation, (4) wax precipitation at low ambient temperatures, (5) bypass failure from filter media collapse, and (6) failure of housing seals with unfiltered bypass flow.

These failure modes have distinct sensor readings across the pressure differential, flow rate, injector return flow and fuel temperature, allowing ML classification when a sufficient volume of training data is available for each of the six classes.

IV. DATASET AND EXPERIMENTAL METHODOLOGY

A. Data Collection Protocol

We gathered sensor data from 42 commercial diesel vehicles (18 light commercial trucks (3.5-12 tonnes GVW), 16 heavy-duty road haulage vehicles (>25 tonnes GVW) and 8 agricultural tractors (75-130 hp) operated by three fleet operators in the Indian states of Punjab, Haryana, and Rajasthan over 18 months (January 2023 - June 2024). The vehicles were fitted with BS-VI compliant aftertreatment systems. All vehicles were fitted with 1 Hz OBD-II data loggers, with additional pressure transducers and thermocouple pairs at the DPF inlet and outlet.

DPF clogging fault labels were determined through a combination of dashboard event logs, workshop inspection results (filter differential pressure at service visits), and gravimetric filter weighing (at service intervals). Labels for fuel filter faults were determined by laboratory testing of fuel filter samples at service intervals, and cross-checked with spectrometric fuel analysis. The data set consisted of around 2.8 million 1-Hz observations following preprocessing, corresponding to 1,890 filter-day operating time across the 42 vehicles.

B. Feature Engineering

Eight main features were constructed from the raw sensor data after resampling to 1-minute chunks (mean, standard deviation and change over 60 seconds):

1. Differential Pressure (ΔP) - upstream - downstream pressure (kPa), soot loading indicator
2. Exhaust Gas Temperature Differential (ΔT) (upstream - downstream EGT (°C)
   regeneration activity indicator
3. Rate of Change of Engine Load - rate of change of the normalized torque, duty cycle indicator
4. Fuel Rate of Change - fuel flow (L/hr)
5. Soot Accumulation Index (SAI) - model estimated soot mass (g/L) from pressure-temperature
6. Coolant Temperature - engine temperature proxy (°C)
7. Vehicle Speed - duty cycle class identifier (km/hr)
8. Regeneration Event Flag - binary signal for regeneration event status

Feature importance analysis using mutual information ranking and recursive feature elimination (RFE) with 5fold cross-validation identified ΔP, ΔT, SAI, and engine load rate of change as the four highest-information features across both modeling tasks. All features were normalized to [0,1] using training-set statistics exclusively, with test set normalization applied using the training-set parameters to prevent data leakage.

C. Train/Test Split and Validation Protocol

An 80/20 vehicle-stratified split was used: 34 vehicles were used for training and 8 vehicles for held-out (test) validation. This stratification, as opposed to random time-series split, ensures test accuracy measures the ability to generalise to new operational conditions rather than simply interpolate between data from known vehicle behaviour [11], which is crucial for fleet PdM. We held out 10% of the training data for early stopping and hyperparameter tuning. For the time-series models (LSTM, CNN-LSTM), sequences were formed by rolling 60-minute windows with 10-minute step.

V. MACHINE LEARNING MODELS AND EXPERIMENTAL SETUP

A. Random Forest

A collection of 300 decision trees trained on random subsets of features, with a maximum depth of 20 and minimum number of samples per leaf of 5. The Random

Forest is the central interpretable benchmark model, with feature importance scores used to help engineer validations of model predictions. Hyperparameters were determined by 5-fold crossvalidated grid search on the training set.

B. XGBoost

Gradient boosted decision trees with 500 trees, learning rate 0.05, tree maximum depth 6 and regularisation weight $\lambda = 1.0$. To avoid overfitting, a subsampling rate of 0.8 for both instances and features was used. XGBoost's ability to handle missing values and its success on structured tabular data for classification tasks made it the front-runner for fuel filter fault classification, where the input is a feature vector rather than a time series.

C. Support Vector Machine

An SVM with a radial basis function (RBF) kernel using $C = 10$ and $\gamma = 0.01$, found by grid search with 5-fold cross-validation. SVM is the traditional maximum margin method and offers a benchmark to compare the improvements enabled by ensemble and deep learning approaches.

D. Long Short-Term Memory Network

A bidirectional LSTM network with a stack of two fully connected LSTM layers, each having 128 memory units, and 60-minute input window. Dropout (0.3) was used between layers to prevent overfitting of the network to the small vehicle-day dataset. Training was performed with the Adam optimizer, initial learning rate 1×10^{-3}, reducing the learning rate when a plateau is reached on the validation loss (factor 0.5, patience 5 epochs), and early stopping (patience 15 epochs). LSTM's ability to capture long-term dependencies in the degradation time series makes it a natural fit for soot build-up accumulation.

E. Hybrid CNN-LSTM

One 1D convolutional layer with 64 filters, kernel size 5 followed by a single LSTM layer of 128 units, then a fully connected classification head with dropout 0.4. The convolutional layer captures local temporal features (transients due to high load, fast temperature fluctuations, regeneration events) and the LSTM provides a sequence-level representation of the cumulative degradation process. This temporal multi-scale architecture was predicted to best capture the dual timescale nature of DPF soot loading: rapid transients due to duty cycle and long-term accumulation of soot due to total time of operation.

VI. SYSTEM ARCHITECTURE FOR INTELLIGENT FILTER MONITORING

A. Component Overview

Our intelligent filter monitoring system has five components that are designed to be integrated into a commercial fleet telematics system: the Edge Inference Node, the Telemetry Aggregation Layer, the Cloud Analytics Engine, the Maintenance Decision Interface and the Model Update Pipeline. These components cooperate in an edge-cloud system that provides edge inference to perform real-time monitoring at the vehicle level and cloud inference to improve the model at the fleet level.

TABLE I. SYSTEM COMPONENT SUMMARY

Component	Primary Function	Deployment Location
Edge Inference Node	Real-time sensor fusion and ML inference	Vehicle ECU / OBD module
Telemetry Aggregation Layer	Batch upload of sensor logs and inference results	Fleet gateway server
Cloud Analytics Engine	Model training, retraining, and fleet analytics	Cloud infrastructure
Maintenance Decision Interface	Alert generation and maintenance scheduling	Fleet management portal
Model Update Pipeline	Incremental model updates pushed to edge nodes	Cloud-toedge OTA

B. Edge Inference Node

The Edge Inference Node is executed on a low-power ARM Cortex-M7 microcontroller on-board the OBD telematics device. It runs quantized versions of the trained ML models - an INT8-quantizedXGBoost for fuel filter fault classification and a pruned LSTM for estimating soot load on the DPF - at 1minute sampling rate. Quantization and pruning shrink the size of the models by 75% and 60% respectively, compared with the full-precision models trained, allowing inference within the 256 KB RAM budget of the target hardware with minimal impact on accuracy (< 0.8% reduction in accuracy after quantization is applied).

The node produces three continuous outputs: a DPF health score (continuous, 0-1), a fuel filter fault class probability score (six-class softmax), and a regeneration optimization recommendation (timing and mode). The outputs are logged locally and sent to the Telemetry Aggregation Layer at user-defined time intervals.

C. Maintenance Decision Interface

The Maintenance Decision Interface converts the output of the ML models into maintenance recommendations for fleet managers. The continuous ML scores have a rule-based alert layer: when the DPF health score drops below a user-defined threshold (θ_DPF, default 0.35) for three inference cycles, a Level 1 alert recommends forced active regeneration within 24 hours. If the score is below 0.15, it triggers a Level 2 alert with a recommended workshop inspection. When probabilities of non-normal fuel filter faults are greater than 0.70, a fault-specific alert is issued with a fault code, life remaining and a maintenance action for a given fault based on a maintenance decision tree.

VII. RESULTS

A. DPF Clog Prediction Performance

Table II shows the complete model performance of predicting DPF clog on the test data with 8 vehicles. The models were trained and tested on the same data with the same preprocessing.

TABLE II. DPF CLOG PREDICTION — MODEL PERFORMANCE (TEST SET, n=8 VEHICLES)

#	Model	Acc.	F1	Recall	AUC
1	CNN-LSTM	94.3%	0.937	0.941	0.978
2	LSTM	91.7%	0.912	0.918	0.961
3	XGBoost	88.4%	0.879	0.875	0.944
4	Rand. Forest	85.1%	0.847	0.843	0.928
5	SVM	79.6%	0.791	0.784	0.876

B. Fuel Filter Fault Classification Performance

Table III shows model performance for the fuel filter fault classification task. The relative performance is distinct from the DPF task, with ensemble models performing better than temporal models.

TABLE III. FUEL FILTER FAULT CLASSIFICATION — MODEL PERFORMANCE (SIX-CLASS)

#	Model	Acc.	F1 (macro)	Recall	AUC
1	XGBoost	91.8%	0.914	0.907	0.967
2	CNN-LSTM	89.5%	0.891	0.886	0.958
3	Rand. Forest	86.3%	0.859	0.855	0.941
4	LSTM	83.7%	0.832	0.826	0.921
5	SVM	76.4%	0.758	0.752	0.889

C. Per-Class Fault Classification Analysis

The two best individual class classification accuracy results were for water ingress (97.2%) and wax crystallization (96.8%) due to their unique pressure signatures that are strongly correlated with temperature, and distinct in feature space. The hardest to classify was the filter bypass failure class with 84.1% classification accuracy, due to its intermittent nature and similarity with severe sediment blockage in the pressure differential feature. The classification accuracy for microbial contamination was 89.3% and was enabled by a distinctive pattern of steadily rising pressure differential with abnormal oscillations in flow rate, which the XGBoost model learned as a high-importance feature combination.

D. Maintenance Scheduling Impact

Integration of the CNN-LSTM DPF model into the predictive maintenance scheduling simulation for the 8 test vehicles resulted in the following changes as compared to the fleet's status quo 15,000 km service interval: a 34.2% reduction in the number of unnecessary filter replacements (from an average of 3.1 to 2.0 replacements per vehicle-year in the test cohort); a 61.4% reduction in unplanned downtime events due to DPF over-loading or filter bypass failure; and a 47.3-hour mean (standard deviation 8.2 hours) advance warning window between the ML-generated alert and the threshold breach. The 47-hour lead time is important in practice because it falls within the fleet workshop scheduling window, allowing maintenance to be planned to account for a rest period.

VIII. DISCUSSION

A. Why CNN-LSTM Leads for DPF Clog Prediction

CNN-LSTM's strong performance on the DPF task is due to the multi-scale temporal dynamics of soot loading. DPF clogging is caused by two processes: high-frequency, high-amplitude transients from hard acceleration and high-load urban driving, causing rapid soot accumulation, and low-frequency, monotonic accumulation over the day and week. Neither network structure alone is ideal for both scales.The LSTM (without convolution) fails to efficiently extract the local transients because its memory is well-adapted to extract long-term trends; the CNN (without recurrence) cannot capture the cumulative trajectory of the DPF because it has no memory beyond the kernel size. ilter's kernel width.

The hybrid architecture essentially decomposes the problem: the convolutional network extracts a collection of local temporal pattern features (hard-acceleration events, period of high load, regeneration heat signatures) which are used as a compressed feature sequence by the LSTM and then encoded into a trajectory characteristic of cumulative filter state. This decomposition is not only empirically successful, but it also corresponds to the physics of the DPF loading process, and offers a level of interpretability that pure end-to-end models do not.

B. XGBoost's Advantage in Fault Classification

The different rank ordering of the algorithms for fuel filter fault classification (with XGBoost as the best and the temporal models as second and fourth) can be explained by the different statistical nature of the two problems. DPF clog prediction is a time series forecasting problem: the input is a time series and the output is a function of its history. Fuel filter fault classification is a structured pattern recognition problem: the different fault types create a static pattern in the feature space that exists at any point in time during the fault, regardless of the past history. Time is a source of noise in this problem.

Gradient-boosted trees are well-adapted to structured pattern recognition because they learn hierarchical rules of feature interaction that effectively separate classconditional feature distributions. For instance, the water ingress class is defined by a particular conjunction of features with high fuel temperature, low differential pressure (water has lower viscosity) and oscillating flow rate; the tree structure of XGBoost automatically learns this conjunction, while the LSTM must learn it via the temporal dynamics at additional time cost and with a higher risk of overfitting on the minority classes.

C. Comparison with Published Literature

The CNN-LSTM result of 94.3% for predicting DPF replacement is better than Gao et al.'s [8] 89.7% using Bayesian neural networks (BNN), probably because of the larger and more diverse data set used here, which better covers the distribution of operating conditions in typical use. The 91.8% XGBoost fuel filter classification accuracy is in line with the 93.1% reported by Wang et al. [9] for a similar task, with a small difference likely due to the six-class taxonomy used in this study (including more sub-classes) versus a four-class taxonomy in Wang et al.'s experiment - a more challenging task with more confusion between classes.

The 34% improvement in unnecessary replacements and 61% improvement in unplanned downtime are significantly greater than the 15-25% and 30-40% ranges reported in other fleet PdM studies [10], [11]. This may be explained by the inefficiency of the fixedinterval maintenance protocol used by the fleet operators in this study, which had not been adjusted to account for changed soot levels due to fuel changes from BS-IV to BS-VI during the study period (the original fixed intervals were set based on BS-IV fuels).

IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

A. Dataset Scope and Generalizability

The data are site specific to northern India, where climate, driving conditions, fuel sulfur content and vehicle operating conditions may vary considerably from fleet operations in Europe, North America or Southeast Asia. The 18-month duration of the study may not have fully captured the seasonal variability of wax crystallization events or accumulation of ash in the filter. The study included only vehicles operating on low-sulfur, BS-VI compliant fuel; performance on older BS-IV fleets or in markets with higher fuel sulfur content may be different because sulfate particulates from fuel sulfur have different DPF loading characteristics compared to soot.

B. Soot Accumulation Index Reliability

The Soot Accumulation Index feature used in the model is an estimation based on differential pressure and temperature measurements using a simplified wallflow filtration equation, rather than the direct gravimetric measurement. This estimation error is systematic and of order proportional to the difference between model parameters and the true filter and soot properties, which is a function of the filter's state of ageing and fuel quality. This error should be assessed in future studies using periodic gravimetric measurements to determine the impact on model accuracy over the filter life.

C. Data Privacy in Fleet Telematics

The real-time tracking of vehicle location, speed, load, and operating patterns that is required for the PdM approach has obvious privacy implications for vehicle owners and drivers. Fleet telematics data has come under renewed scrutiny under data protection laws, and its application for predictive maintenance must be subject to data minimization policies: only the sensor data streams necessary for filter health inference should be stored; raw operational data should not be stored beyond the inference window. Training data for the model should be governed by suitable data governance agreementswith fleet operators, with explicit provisions for data deletion rights.

X. FUTURE RESEARCH DIRECTIONS

A. Digital Twin Integration

The most promising short-term follow up to this work is to couple the ML-based filter health model to a physics-based digital twin of the entire diesel aftertreatment system. A digital twin integrates the pattern recognition ability of the ML model with a physics-informed simulation of the DPF, DOC and SCR system, and uses the outputs of the ML model as estimates of the state of the system to initialise and update the physics simulation [12]. The combined approach would allow the system to provide physically meaningful predictions, such as ash capacity, optimal DPF regeneration temperature and downstream catalyst degradation, which are beyond the scope of the data-driven model.

B. Federated Fleet Learning

ML models trained on data from 42 vehicles in a single fleet, as part of a research project, will not be good enough for deployment in diverse commercial fleets (multiple makes, multiple use cases). Federated learning approaches that train a global model by aggregating local gradient updates from the fleet edge devices at multiple fleet operator nodes - without sharing raw sensor data - provide a scalable solution to fleet training without violating data privacy [13]. Future research should implement and test a federated learning framework for the filter health prediction models, to compare the accuracy improvement from training across larger and more varied fleets, against the communication costs of gradient synchronization.

C. Multi-Sensor Fusion and Acoustic Diagnostics

The current approach uses electronic sensor streams available via OBD. The next step should be to explore the use of acoustic emission sensors installed on the DPF housing, which have been shown to respond to early onsets of substrate cracking and ash cake delamination that cannot be detected in the pressure-temperature stream [14]. The combination of acoustic features with electronic sensor streams in a multi-modal ML framework could increase the lead time for mechanical failure modes, and increase classification performance for the most critical failure modes: substrate fracture and bypass failure.

D. Transfer Learning Across Engine Families

The training procedure adopted in this work considers each engine family as a different domain, meaning that data for training a model could need to be collected for each new engine family. Domain adaptation and fine-tuning of pre-trained feature extractors (transfer learning) can potentially expedite model deployment for new engine families, by using the learned features from the current data. Future research should explore how much labeled data should be collected in the target domain to reach an acceptable performance level via fine-tuning vs. training from scratch, and which parts of the architecture are most transferable among engine families with different aftertreatment system configurations.

XI. CONCLUSION

This paper has shown that data analytics and machine learning provide a significant and feasible approach to modernise diesel filter maintenance. Using a dataset of 42 commercial vehicles, three fleet owners and 18 months of operation in northern India, we tested five machine learning (ML) architectures and found that architecture-task fit is a key driver of model performance.

The CNN-LSTM hybrid architecture has predicted DPF clogging with 94.3% accuracy and an F1-score of 0.937, surpassing single LSTM, ensemble, and classical (support vector machine) models. This performance can be attributed to its ability to capture both short- and long-term time dependencies, which align with the multiscale nature of soot buildup. XGBoost achieved 91.8% accuracy for fuel filter fault diagnosis across six classes, exceeding the temporal architectures because the fuel filter fault detection task is not a temporal trajectory prediction problem (as is the case with the DPF clog prediction task), but rather a pattern recognition problem.

These models' application in a predictive maintenance scheduling system led to a 34.2% reduction in premature filter replacement and a 61.4% reduction in unplanned downtime events versus the fixed-interval fleet maintenance protocol, with a 47-hour average warning period, which is consistent with fleet maintenance schedules. These findings provide a convincing quantitative justification for ML-based PdM to replace fixed-interval maintenance of diesel filters in commercial fleets.

The system design proposed in this paper - a hybrid edge-cloud system with quantized inference at the edge, telemetry pooling and model updates in the cloud - offers a deployable pattern that overcomes the computational and network limitations of commercial fleet operations. Research on digital twins, federated learning across fleets, sensor fusion and transfer learning across engine families will be needed to apply the approach across the broad range of global diesel fleet operations.

This work is significant for more than just reducing maintenance costs. With tightening emission standards worldwide, the need to keep diesel aftertreatment systems operating in optimal condition is increasingly linked to compliance. An ML-based filter maintenance system that effectively prevents over-loading of the diesel particulate filter (DPF) and fuel system contamination is also a maintenance and emissions compliance system - a dual benefit that motivates investment in its development and scale-up.

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